ETH zürich

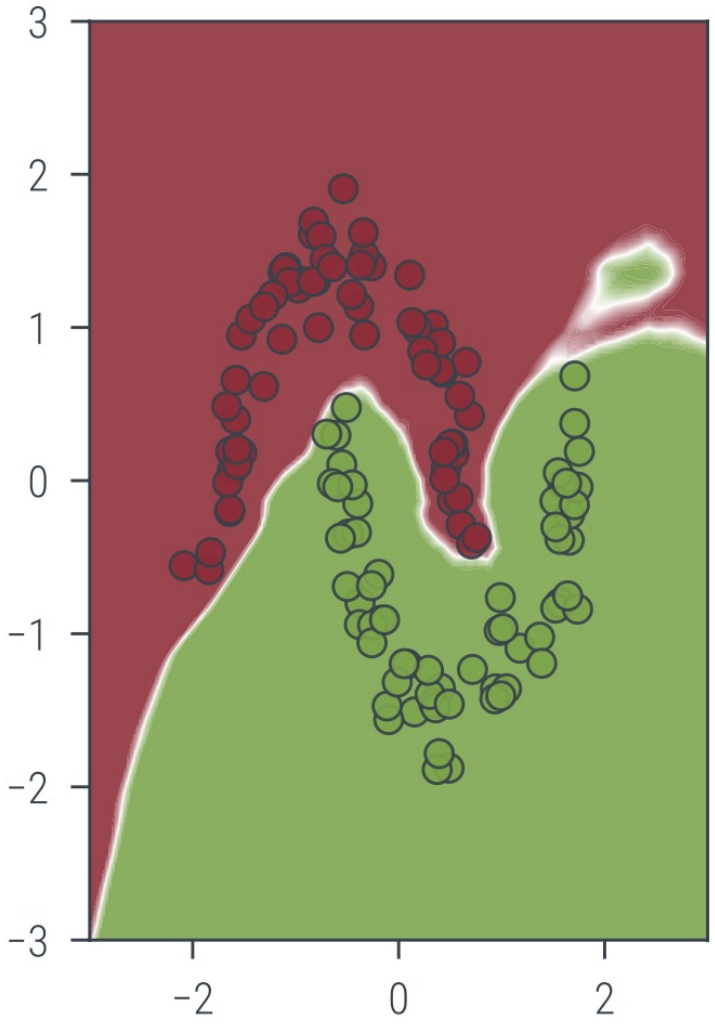
D INFK

Probabilistic Artificial Intelligence

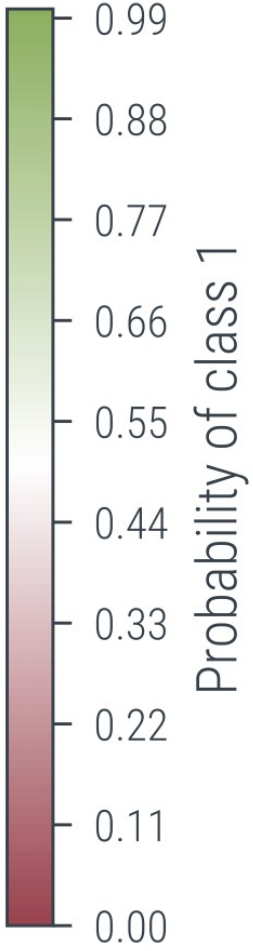
Variational Inference

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Institute for Machine Learning

Motivation: Uncertainty in classification tasks

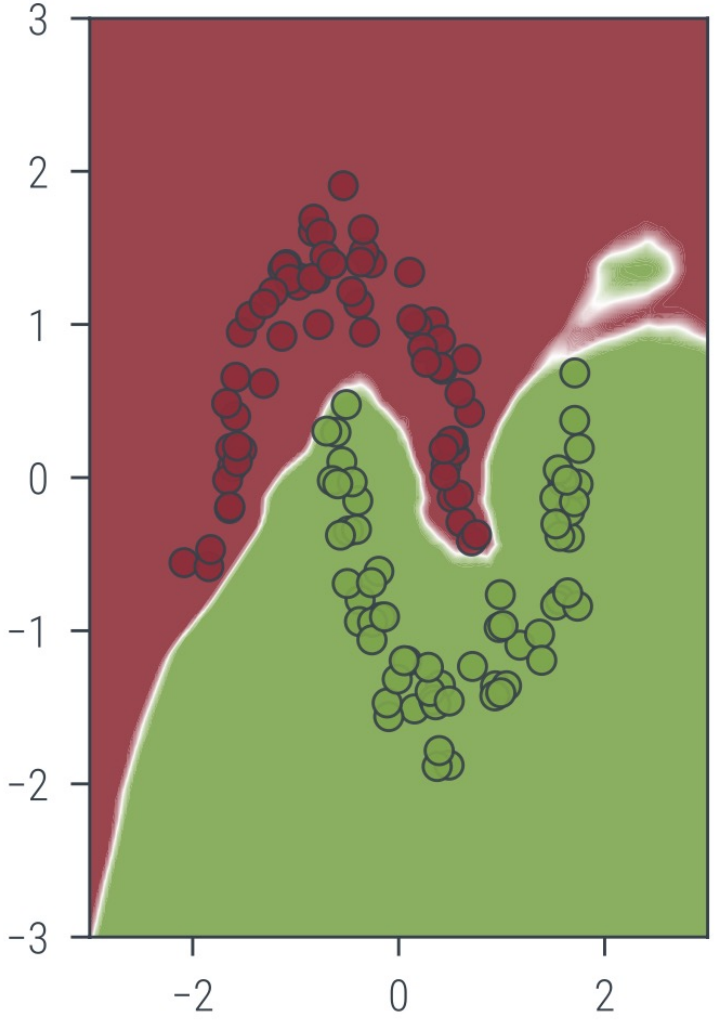


MAP estimate

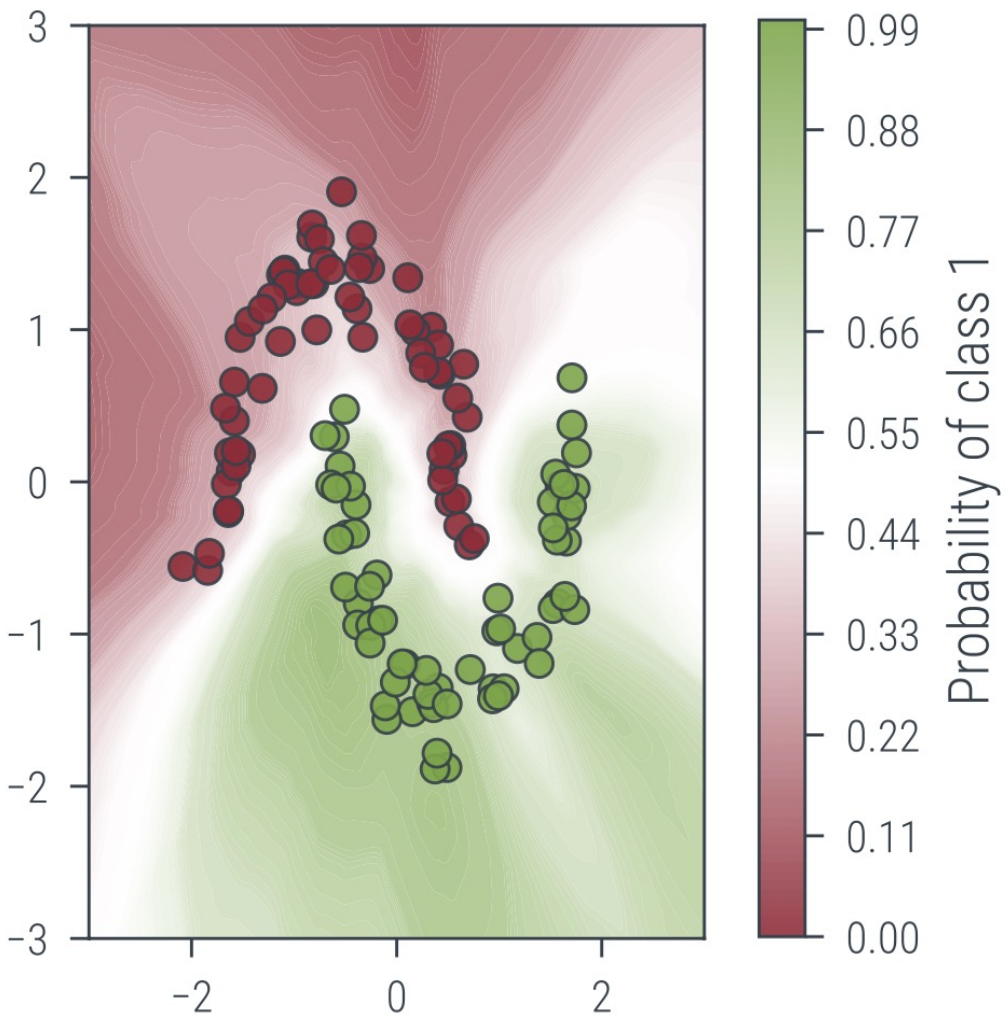


[Philipp Hennig]

Motivation: Uncertainty in classification tasks



MAP estimate



(Approx.) Bayesian estimate

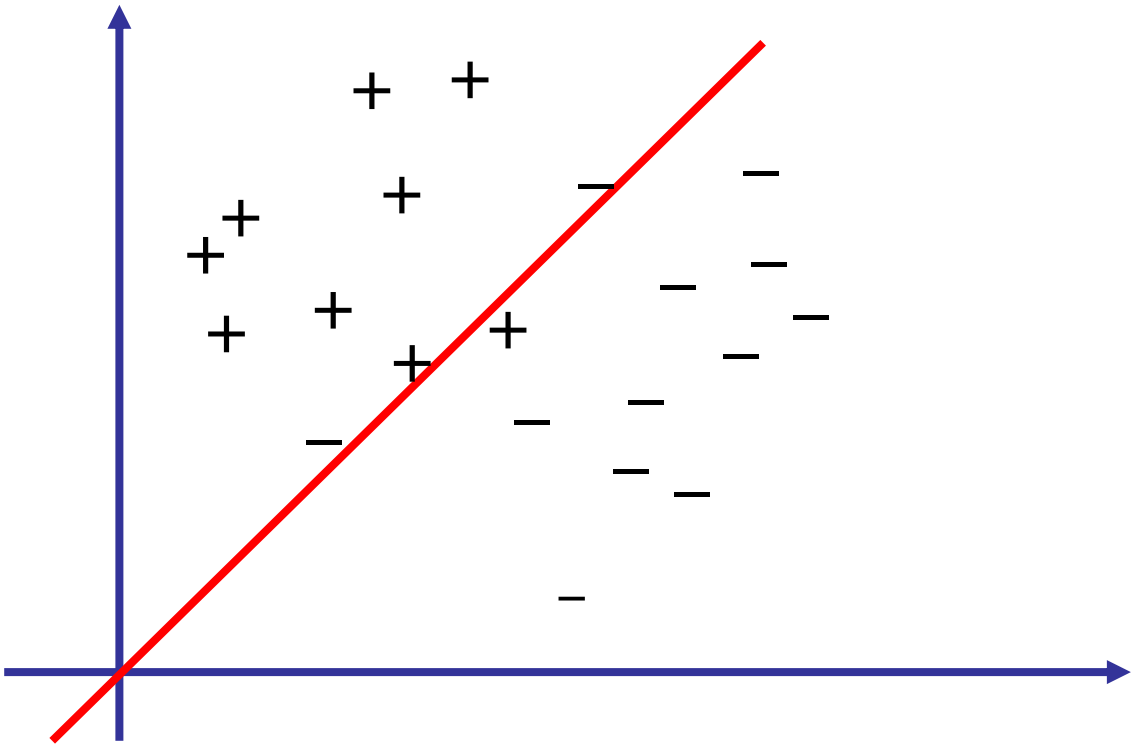
[Philipp Hennig]

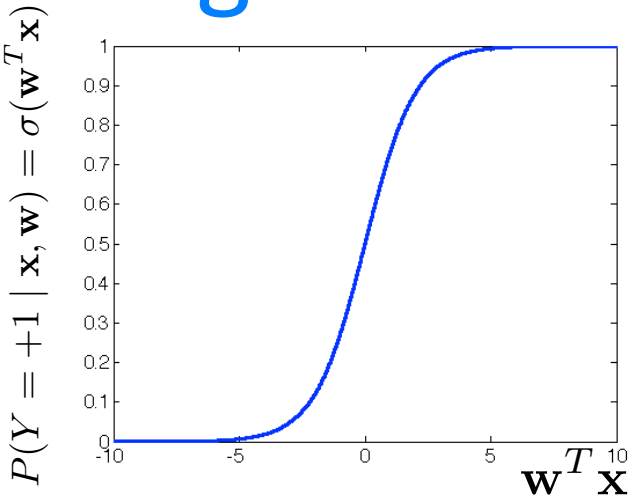
Bayesian learning more generally

- Prior $p(\theta)$
- Likelihood: $p(y_{1:n} \mid x_{1:n}, \theta) = \prod_{i=1}^{n} p(y_i \mid x_i, \theta)$
- Posterior: $p(\theta \mid x_{1:n}, y_{1:n}) = \frac{1}{Z} p(\theta) \prod_{i=1}^{n} p(y_i \mid x_i, \theta)$

  where $Z = \int p(\theta) \prod_{i=1}^{n} p(y_i \mid x_i, \theta) d\theta$
- Predictions: $p(y^* \mid x^*, x_{1:n}, y_{1:n}) = \int p(y^* \mid x^*, \theta) p(\theta \mid x_{1:n}, y_{1:n}) d\theta$
- For Bayesian linear regression and GP regression, these (high-dimensional) integrals are closed-form! ☺
- In general, this is not the case ➔ need approximations

Example: Bayesian logistic regression

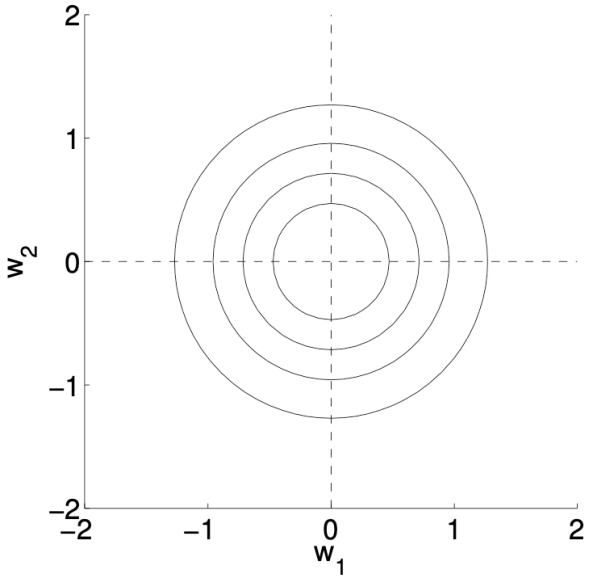


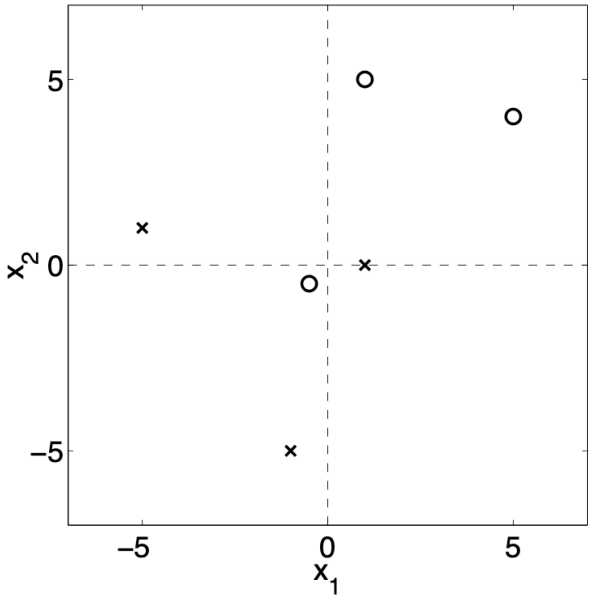


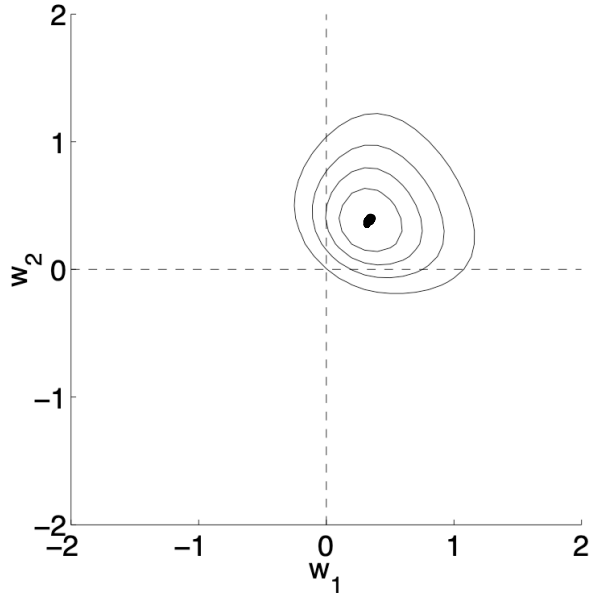
$$\sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{w}^T\mathbf{x})}$$

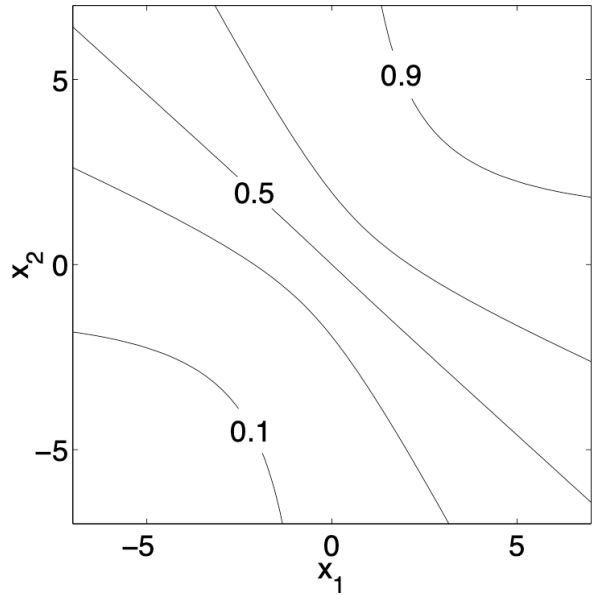
$$P(y \mid \mathbf{x}, \mathbf{w}) = \mathrm{Ber}(y; \sigma(\mathbf{w}^T\mathbf{x}))$$

Bayesian logistic regression [GPML]









Approximate inference

- Will discuss general approaches for performing approximate inference in intractable distributions (i.e., partition function / normalizer hard to compute)

$$p(\theta \mid y) = \frac{1}{Z} p(\theta, y)$$

- Hereby, y are the observations (the data), and θ the latent variables (the parameters)
- We'll assume we can evaluate the joint distribution, but *not* the normalizer $\boldsymbol{Z}$
- Note that we often leave out the inputs $\boldsymbol{x}$ to keep notation simple

General approaches

- Variational inference seeks to approximate the intractable distribution **p** by a simple one **q** that is "as close as possible"

$$p(\theta \mid y) = \frac{1}{Z} p(\theta, y) \approx q(\theta \mid \lambda)$$

- Markov-Chain Monte Carlo methods seek to approximate $\boldsymbol{p}$ by (approximate) samples from $\boldsymbol{p}$ (constructed by simulating a Markov Chain)
  → more information in lecture notes

General approaches

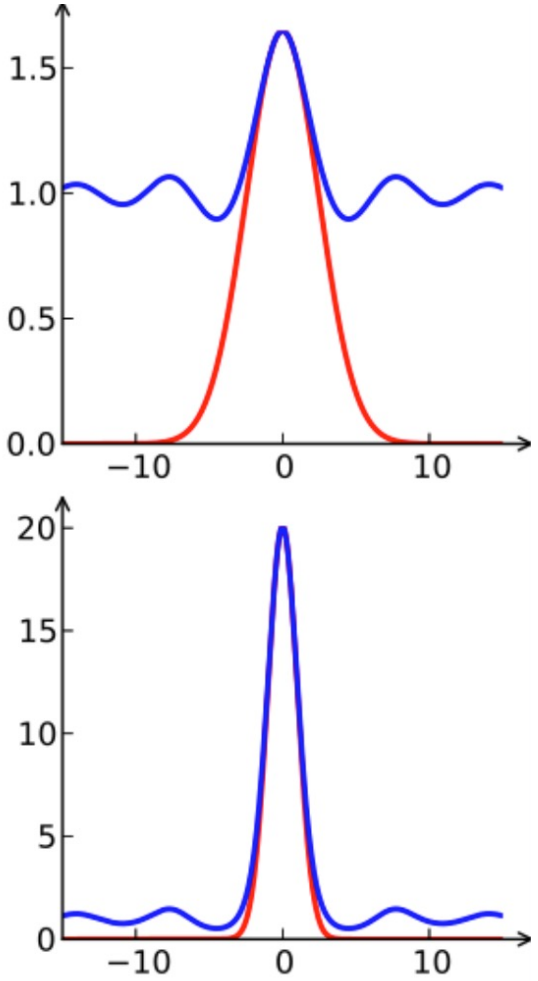
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Laplace approximation

- Laplace approximation:
  Gaussian approximation to the posterior, obtained from a second-order Taylor expansion around the posterior mode



[https://en.wikipedia.org/wiki/Laplace%27s_method]

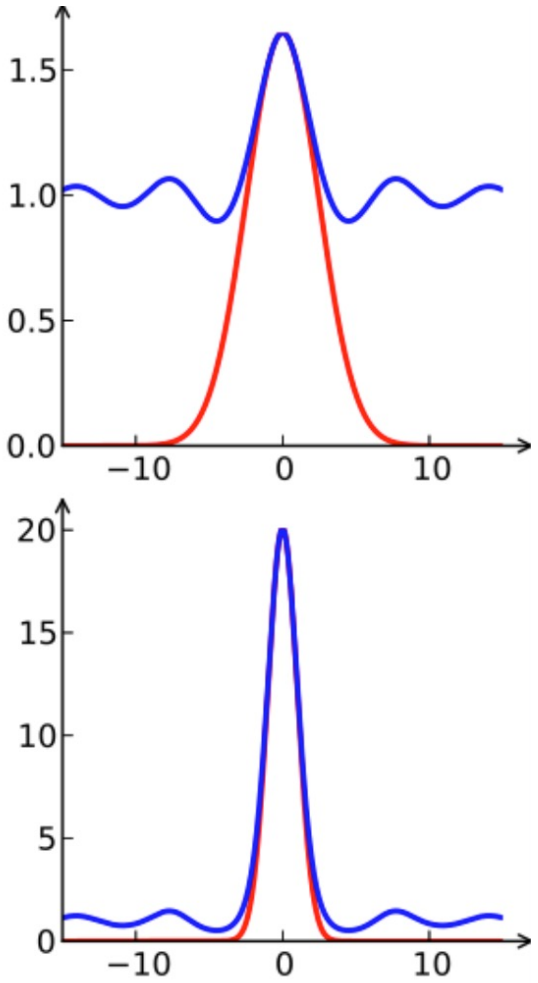
Laplace approximation

- Laplace approximation:
  Gaussian approximation to the posterior, obtained from a second-order Taylor expansion around the posterior mode

$$q(\theta) = \mathcal{N}(\theta; \hat{\theta}, \Lambda^{-1})$$

$$\hat{\theta} = \arg\max_{\theta} p(\theta \mid y)$$

$$\Lambda = -\nabla\nabla \log p(\hat{\theta} \mid y)$$



[https://en.wikipedia.org/wiki/Laplace%27s_method]

Laplace Approx. for Bayesian log. regression

Finding the mode

$$\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} p(\mathbf{w}|\mathbf{x}_{1:n}.y_{1:n})$$

$$\equiv \arg\min_{\mathbf{w}} \sum_{i=1}^{n} \log\Big(1 + \exp(-y_i \mathbf{w}^T \mathbf{x}_i)\Big) + \lambda||\mathbf{w}||_2^2$$

- This is just standard (regularized) logistic regression!
- Can solve, e.g., using stochastic gradient descent (see introduction to ML)
- Don't need to know normalizer *Z*

Recall: Stochastic Gradient Descent

- **Goal**: minimize stochastic objectives

$$L(\theta) := \mathbb{E}_{\mathbf{x}\sim p} l(\theta; \mathbf{x})$$

SGD

- Initialize θ_1
- For t=1 to T
  - Draw minibatch $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}, \mathbf{x}_i \sim p$
  - Update $\theta_{t+1} \leftarrow \theta_t - \eta_t \frac{1}{m} \sum_{i=1}^{m} \nabla_\theta l(\theta_t; \mathbf{x}_i)$

- Many variants (Momentum, AdaGrad, ADAM, …)
- For proper learning rate η_t converges to (local) minimum
- Gradient $\nabla_\theta l(\theta; \mathbf{x})$ often obtained by automatic differentiation

Recall: SGD for logistic regression (m=1)

- Initialize **w**
- For t = 1,2,...
  - Pick data point $(\boldsymbol{x}, y)$ uniformly at random from data D
  - Compute probability of misclassification with current model

$$\hat{P}(Y = -y \mid \boldsymbol{w}, \boldsymbol{x}) = \frac{1}{1 + \exp(y\, \boldsymbol{w}^T \boldsymbol{x})}$$

  - Take gradient step

$$\boldsymbol{w} \leftarrow \boldsymbol{w}\,(1 - 2\lambda\eta_t) + \eta_t\, y\, \boldsymbol{x}\, \hat{P}(Y = -y \mid \boldsymbol{w}, \boldsymbol{x})$$

Finding the covariance

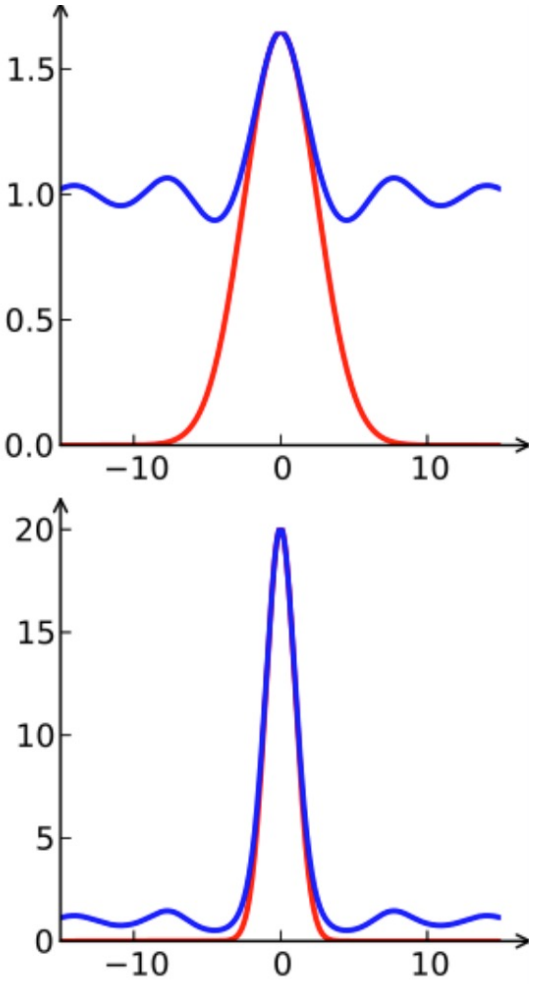
$$\Lambda = -\nabla\nabla \log p(\hat{w} \mid x_{1:n}, y_{1:n})$$

$$= -\nabla\nabla\big(\log p(\hat{w}) + \log\big(p(y_{1:n} \mid \hat{w}, x_{1:n})\big)\big)$$

$$= \sigma_p^{-2}\boldsymbol{I} + \sum_{i=1}^{n} x_i x_i^T \pi_i(1-\pi_i)$$

$$= \sigma_p^{-2}\boldsymbol{I} + \boldsymbol{X}^T \operatorname{diag}([\pi_i(1-\pi_i)]_i)\boldsymbol{X}$$

where $\pi_i = \sigma(\hat{w}^T x_i) = P(\,Y = 1 \mid \hat{w}, x_i\,)$



Crucially, Λ does *not* depend on the normalizer Z

Making predictions

Making predictions

- Given Gaussian approximation of posterior

$$p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n}) \approx q(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \hat{\mathbf{w}}; \Lambda^{-1})$$

the predictive distribution for test point x* is given by

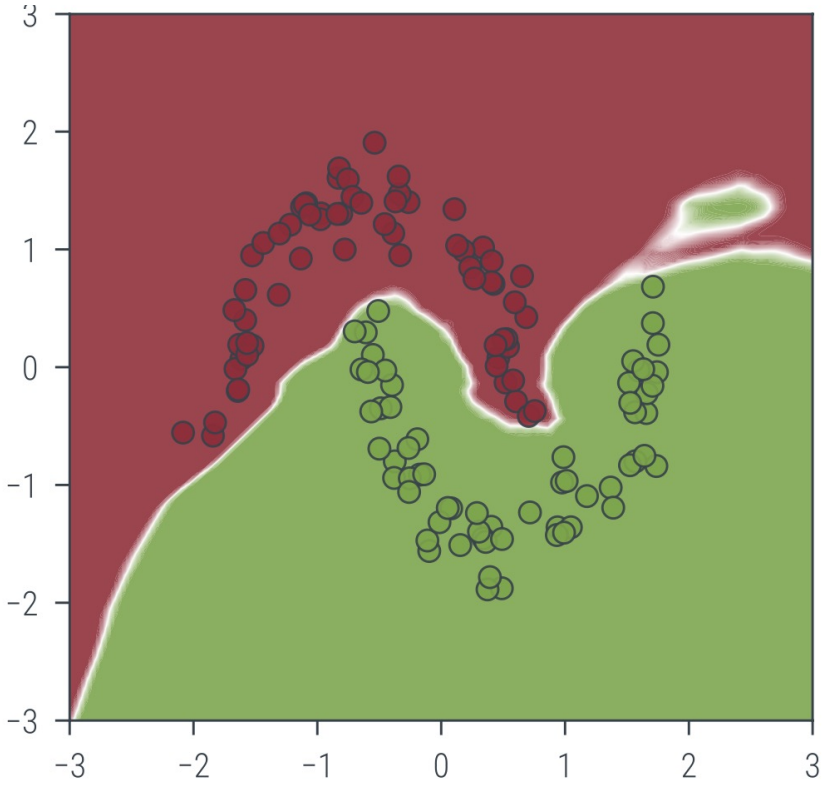
$$p(y^* \mid \mathbf{x}^*, \mathbf{x}_{1:n}, y_{1:n}) \approx \int \sigma(y^* \mathbf{w}^T \mathbf{x}) \mathcal{N}(\mathbf{w}; \hat{\mathbf{w}}, \Lambda^{-1}) d\mathbf{w}$$

$$= \int \sigma(y^* f) \mathcal{N}(f; \hat{\mathbf{w}}^T \mathbf{x}^*, \mathbf{x}^{*T} \Lambda^{-1} \mathbf{x}^*) df$$

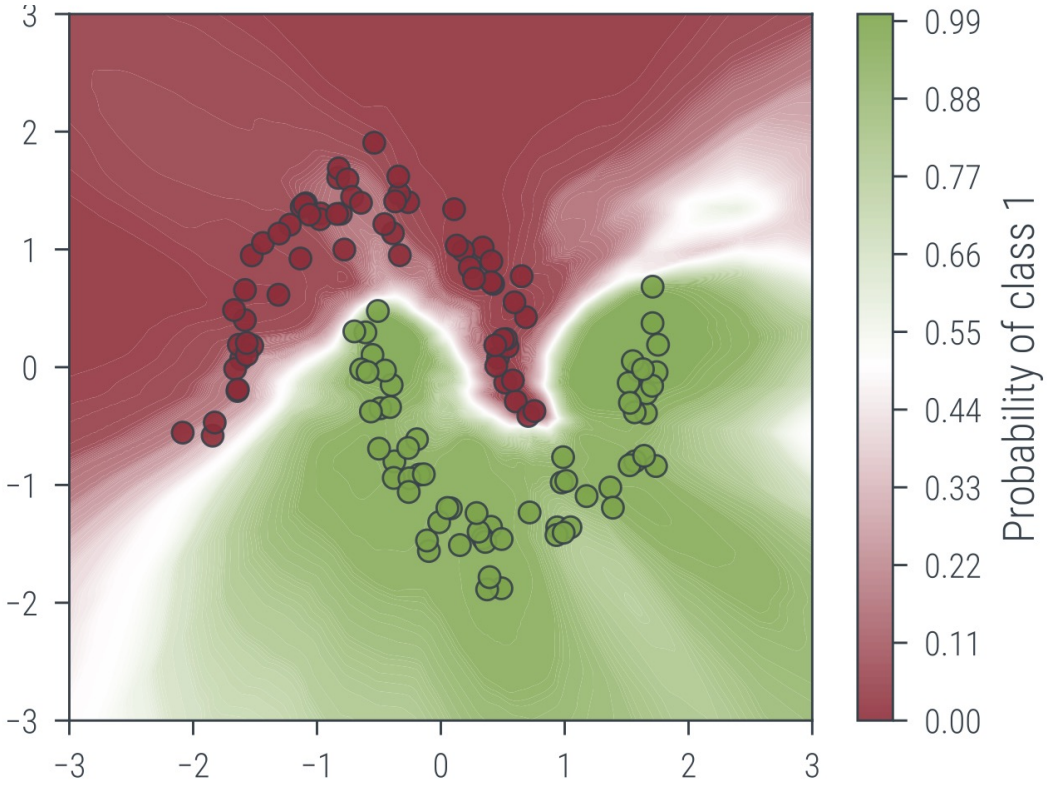
- This one-dimensional integral can be easily approximated efficiently using numerical quadrature
- [Side note: For other link functions (e.g., Gaussian CDF), can even be calculated analytically]

Demo: Laplace approximation

Laplace approximation for classification



MAP

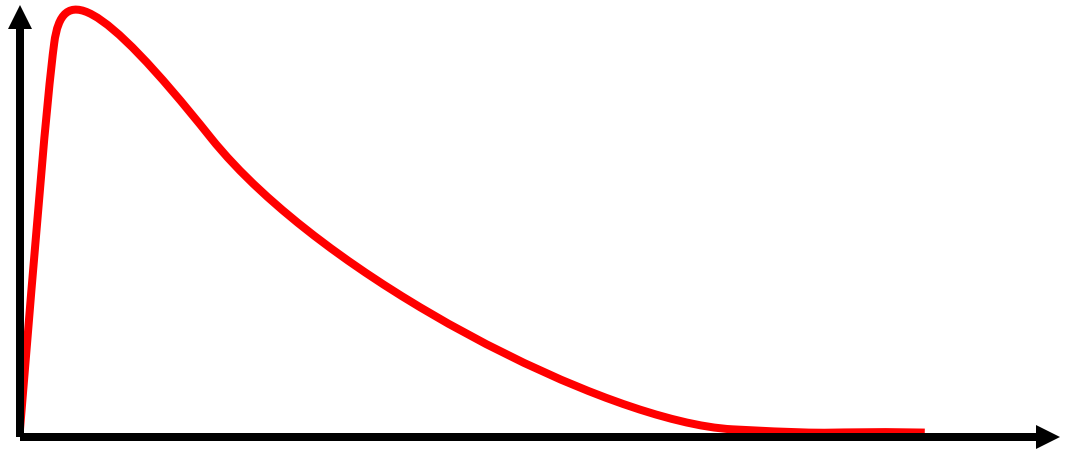


Laplace approximation

[Philipp Hennig]

Issues with Laplace Approximation

- Laplace approximation first greedily seeks the mode, and then matches the curvature
- This can lead to poor (e.g., overconfident) approximations

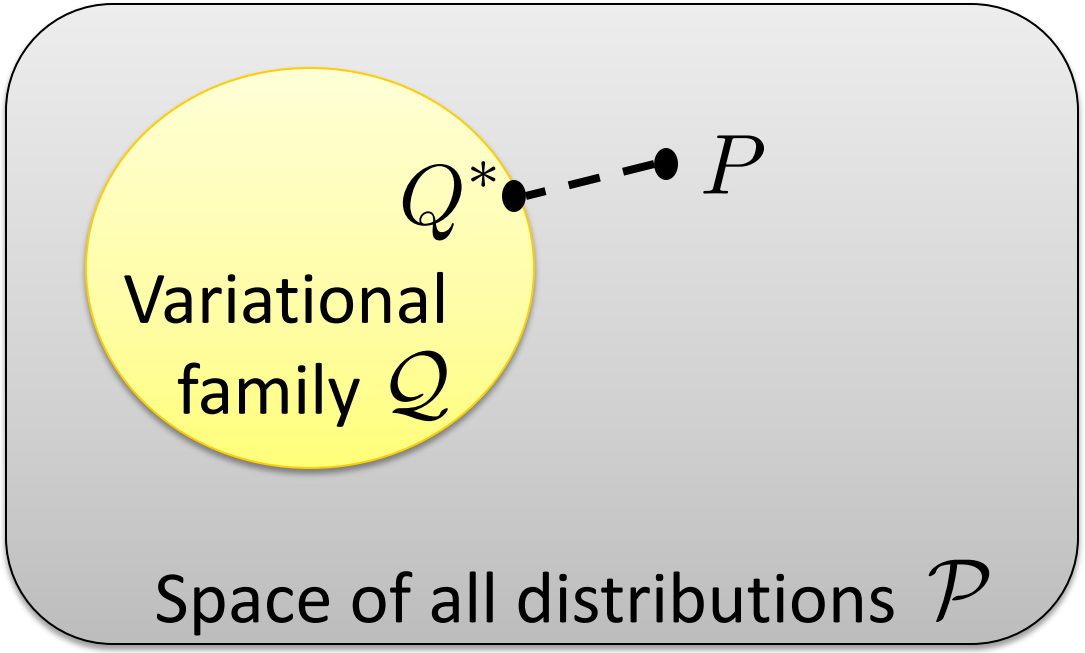


Variational inference

- Given unnormalized distribution

$$p(\theta \mid y) = \frac{1}{Z} p(\theta, y)$$

- Try to find a "simple" (tractable) distribution that approximates *p* well



$$q^* \in \arg\min_{q \in \mathcal{Q}} KL(q||p)$$

Simple distributions

$$q^* \in \arg\min_{q \in \mathcal{Q}} KL(q||p)$$

- Need to specify a variational family (of simple distributions)
- E.g.: Gaussian distributions; Gaussians with diagonal covariance, ….

$$\mathcal{Q} = \{q(\theta) = \mathcal{N}(\theta; \mu, \mathrm{diag}([\sigma^2]))\}$$

KL-Divergence

- Given distributions *q* and *p*, Kullback-Leibler divergence between *q* and *p* is

$$KL(q||p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta)} d\theta$$

- Properties
  - Non-negative
  - Zero if and only if P & Q agree almost everywhere
  - Not generally symmetric

Example: KL divergence between Gaussians

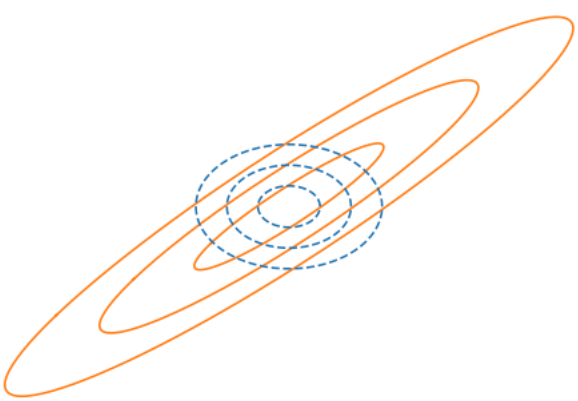
- Consider two Gaussian distributions p and q

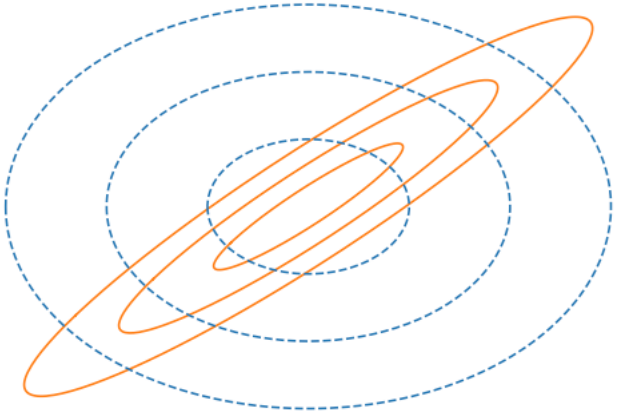
$$p = \mathcal{N}(\mu_0, \Sigma_0),\ q = \mathcal{N}(\mu_1, \Sigma_1)$$

- Then it holds that

$$KL(p||q) = \frac{1}{2}\left(\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^T\Sigma_1^{-1}(\mu_1 - \mu_0) - d + \ln\left(\frac{|\Sigma_1|}{|\Sigma_0|}\right)\right)$$

Illustration of variational approximation





Entropy

- Entropy of a distribution:

$$H(q) = -\int q(\theta) \log q(\theta) d\theta$$

- Entropy of a product distribution: $q(\theta_{1:d}) = \prod_{i=1}^{d} q_i(\theta_i)$

$$H(q) = \sum_{i=1}^{d} H(q_i)$$

- Example: Entropy of a Gaussian

$$H(\mathcal{N}(\mu, \Sigma)) = \frac{1}{2} \ln |2\pi e \Sigma|$$

Minimizing KL Divergence

Minimizing KL Divergence

$$\arg\min_q KL(q||p) = \arg\min_q \int q(\theta) \log \frac{q(\theta)}{\frac{1}{Z}p(\theta, y)} d\theta$$

$$= \arg\max_q \int q(\theta)[\log p(\theta, y) - \log Z - \log q(\theta)]d\theta$$

$$= \arg\max_q \int q(\theta) \log p(\theta, y) d\theta + H(q)$$

$$= \arg\max_q \mathbb{E}_{\theta \sim q(\theta)}[\log p(\theta, y)] + H(q)$$

$$= \arg\max_q \mathbb{E}_{\theta \sim q(\theta)}[\log p(y \mid \theta)] - KL(q||p(\cdot))$$

Maximizing Lower Bound on Evidence

Maximizing Lower Bound on Evidence

$$\log p(y) = \log \int p(y \mid \theta) p(\theta) d\theta$$

$$= \log \int p(y \mid \theta) \frac{p(\theta)}{q(\theta)} q(\theta) d\theta$$

$$= \log \mathbb{E}_{\theta \sim q} \left[p(y \mid \theta) \frac{p(\theta)}{q(\theta)} \right] d\theta$$

$$\geq \mathbb{E}_{\theta \sim q} \left[\log \left(p(y \mid \theta) \frac{p(\theta)}{q(\theta)} \right) \right] d\theta$$

$$= \mathbb{E}_{\theta \sim q} \left[\log p(y \mid \theta) \right] - KL(q || p(\cdot))$$

Inference as Optimization

- Thus,

$$\arg\min_{q\in\mathcal{Q}} KL(q||p(\cdot \mid y)) = \arg\max_{q\in\mathcal{Q}} \mathbb{E}_{\theta\sim q}\Big[\log p(y \mid \theta)\Big] - KL(q||p(\cdot))$$

$$= \arg\max_{q\in\mathcal{Q}} \underbrace{\mathbb{E}_{\theta\sim q}\Big[\log p(y,\theta)\Big] + H(q)}$$

"ELBO" (Evidence lower bound) *L(q)*

- Thus, prefer distributions *q* that maximize the expected (joint/conditional) data likelihood, but are also uncertain / close to the prior

ELBO for Bayesian logistic regression

$$L(\lambda) = \mathbb{E}_{\theta \sim q(\cdot|\lambda)} \Big[\log p(y \mid \theta) \Big] - KL(q_\lambda || p(\cdot))$$

Gradient of the ELBO

$$\nabla_\lambda L(\lambda) = \nabla_\lambda \left[\mathbb{E}_{\theta \sim q(\cdot|\lambda)}[\log p(y, \theta)] + H\big(q(\cdot|\,\lambda)\big) \right]$$

$$= \nabla_\lambda \left[\mathbb{E}_{\theta \sim q(\cdot|\lambda)}[\log p(y \mid \theta)] - KL\big(q(\cdot|\,\lambda) || p(\cdot)\big) \right]$$

- Need to differentiate an expectation w.r.t. *q*
- Unfortunately *q* depends on the variational params.

- Key idea: Rewrite in a way that allows Monte Carlo approximation. Different approaches
  - Score gradients (not discussed further here)
  - Reparametrization gradients

Reparametrization Trick

- Suppose we want to calculate

$$\nabla_\lambda \mathbb{E}_{\theta \sim q_\lambda}[f(\theta)]$$

- **Idea**: find random variable $\epsilon \sim \phi$ **not dependent** on λ s.t. $\theta = g(\epsilon; \lambda)$. Then it holds (under some conditions*):

$$\mathbb{E}_{\theta \sim q_\lambda}[f(\theta)] = \mathbb{E}_{\epsilon \sim \phi}\left[f\left(g(\epsilon, \lambda)\right)\right]$$

- Thus

$$\begin{aligned}\nabla_\lambda \mathbb{E}_{\theta \sim q_\lambda}[f(\theta)] &= \nabla_\lambda \mathbb{E}_{\epsilon \sim \phi}\left[f\left(g(\epsilon, \lambda)\right)\right] \\ &= \mathbb{E}_{\epsilon \sim \phi}\left[\nabla_\lambda f\left(g(\epsilon, \lambda)\right)\right]\end{aligned}$$

*see notes

Example: Gaussians

- Suppose we use a Gaussian variational approximation

$$q_\lambda(\theta) = \mathcal{N}(\theta; \mu, \Sigma)$$

- Can reparametrize $\theta = C\epsilon + \mu$ such that $\Sigma = CC^T$ and $\phi(\epsilon) = \mathcal{N}(\epsilon; 0, I)$

- Can w.l.o.g. choose *C* to be lower-diagonal

Reparametrizing the ELBO for B. Log. Reg.

$$\nabla_\lambda L(\lambda) = \nabla_\lambda \mathbb{E}_{\theta \sim q_\lambda}[\log p(\, y \mid \theta\,)] - \nabla_\lambda KL(q_\lambda || p(\cdot))$$

$$= \nabla_{C,\mu} \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[\log p(y \mid C\epsilon + \mu)] - \nabla_{C,\mu} KL(q_{C,\mu} || p(\cdot))$$

- Can compute $\nabla_{C,\mu} KL(q_{C,\mu} || p(\cdot))$ exactly (e.g., auto-diff)
- Can obtain unbiased stochastic gradient estimate of

$$\nabla_{C,\mu} \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[\log p(y \mid C\epsilon + \mu)]$$

Black Box Stochastic Variational Inference

- Maximizing the ELBO using stochastic optimization (e.g., Stochastic Gradient Ascent)
- Can obtain unbiased gradient estimates, e.g., via reparameterization trick, or score gradients:

$$\nabla_\lambda L(\lambda) = \mathbb{E}_{\theta \sim q_\lambda}[\nabla_\lambda \log q_\lambda(\theta) \, (\log p(y, \theta) - \log q_\lambda(\theta)) \,]$$

- For diagonal q, only "twice as expensive" as MAP inference
- Only need to be able to differentiate the (unnormalized) joint probability density *p* and *q*
- **Outlook**: Can achieve better performance, e.g., using
  - Natural gradients
  - Variance reduction techniques (e.g., control variates)

Demo: Bayesian logistic regression

Side Note: Gaussian Process Classification

- All our discussions naturally generalize from Bayesian linear regression to Gaussian process classification: $P(f) = GP(\mu, k)$ and $P(y \mid f, x) = \mathrm{Ber}\big(y; \sigma(f(x))\big)$

- Often implemented using pseudo inputs, and maximizing the ELBO

$$\sum_{i=1:n} \mathbb{E}_{q(f_i)}[\log(p(y_i \mid f_i)] - KL(q(\boldsymbol{u})||p(\boldsymbol{u}))$$

where $q(f_i) \coloneqq \int p(f_i \mid \boldsymbol{u}) q(\boldsymbol{u}) d\boldsymbol{u}$

Variational Inference Summary

- Variational inference reduces inference ("summation/integration") to optimization
- Can use highly efficient stochastic optimization techniques to find approximations
- Quality of approximation hard to analyze

Reading

- GPML Chapter 3 (3.1-3.4)

- Black Box Variational Inference
  Ranganath, Gerrish, Blei
  http://proceedings.mlr.press/v33/ranganath14.pdf
- Stochastic Variational Inference using reparametrization
  Titsias, Lázaro-Gredilla
  http://proceedings.mlr.press/v32/titsias14.pdf
- Scalable Variational Gaussian Process Classification
  Hensman, Matthews, Ghahramani
  http://proceedings.mlr.press/v38/hensman15.pdf
- Survey on Monte Carlo Gradient Estimation
  Mohamed, Rosca, Figurnov, Mnih
  https://www.jmlr.org/papers/volume21/19-346/19-346.pdf
- Black-Box Stochastic Variational Inference in Five Lines of Python
  Duvenaud, Adams
  https://www.cs.toronto.edu/~duvenaud/papers/blackbox.pdf